

Deep Space Communication utilizing IP Stack and Reinforcement Learning Algorithms

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Introduction & Motivation

- Adopting an IP-centric approach leverages standardized and scalable internet protocols (e.g., IPv6) that are already proven on Earth to improve deep-space communications.
- Existing RF methods face challenges such as high latency and limited interoperability, which are increasingly problematic for interplanetary links.
- Incorporating reinforcement learning into delay-tolerant networking (DTN) enables dynamic, optimal routing decisions in environments with long delays and variable conditions.
- Our aim is a future-proof deep-space communication framework.

Research Methodology

- Ka-Band Parabolic Reflector Antenna implemented with MATLAB's Antenna Designer toolbox. (See Table I)
- Link Case Studies (Fig. 2):
 Case A: Earth-Moon direct link.
 Case B: Earth-Mars direct link.
 Case C: Moon-Mars direct link.
 Emphasis on direct satellite-to-satellite communication.
- Performance Analysis:
 Theoretical link budget analysis is validated via monte carlo simulations.
 SCPS/IPv6 is used to demonstrate parsing of a small test message.
 Comparative analysis is performed for SCPS-TP and RL-DTN implementation.

Antenna Type	Parabolic Reflector
Frequency	32 to 34.2 GHz
Diameter	1.5 m
Focal Length	0.7 m
f/d ratio	0.467 m
Exciter	Conical horn
Aperture radius of the exciter	0.102 m

Table I: Antenna Specifications

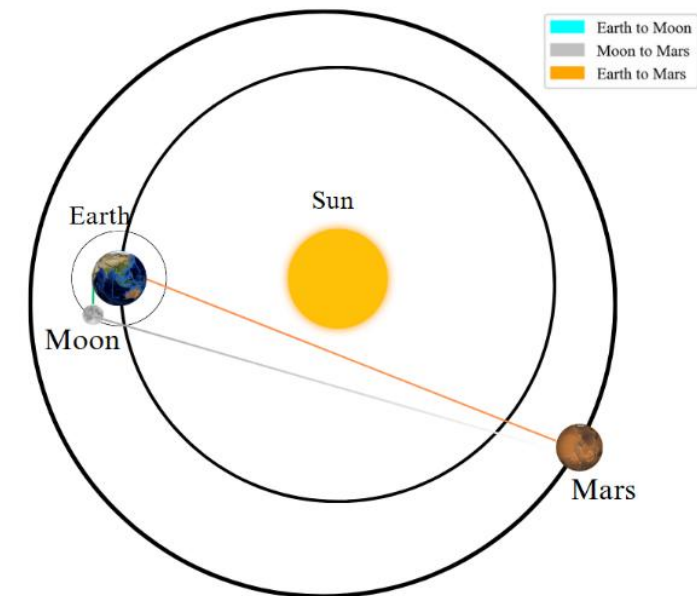


Fig. 2: Interplanetary Links

Research Methodology

- The link budget analysis provides vital information on how much power the system will require.
- Table II present the theoretical link budget analysis carried out based on the data from the antenna [6, 11].
- The received power is well within the acceptable threshold for all the three cases.

Parameters	Earth to Moon		Earth to Mars		Moon to Mars	
	Up-Link	Down-Link	Up-Link	Down-Link	Up-Link	Down-Link
Distance (m)	3.84×10^8	3.84×10^8	4.00×10^{11}	4.00×10^{11}	4.014×10^{10}	4.014×10^{10}
Frequency (GHz)	34.2	32	34.2	32	34.2	32
FSPL (dB)	234.81	234.23	277.43	294.59	275.20	274.59
Data Rate (Mbps)	4.5	4.5	3	3	4.5	4.5
Transmitter Power (dBm)	73	50	73	50	50	60
Transmitter Gain (dBi)	47	46.7	47	46.7	47	46.7
Receiver Gain (dBi)	47	47.6	47	47.6	47	47.6
Losses (dB)	4	4	2	2.71	4	4
Receiver Power (dBm)	-79.21	-101.33	-112.43	-153.19	-135.20	-124.89

Table II: Link Budget Parameters

Results & Discussion

- Monte Carlo Simulation models randomness in signal propagation.
- Introduces variations in signal attenuation, noise and path loss.
- Validates theoretical link budget through multiple trials.
- Simulated values closely follow the theoretical curve, showing high correlation.
- Case A: stable & predictable link, minimal variations.
- Case B & C: Greater signal fluctuations due to the greater distance.

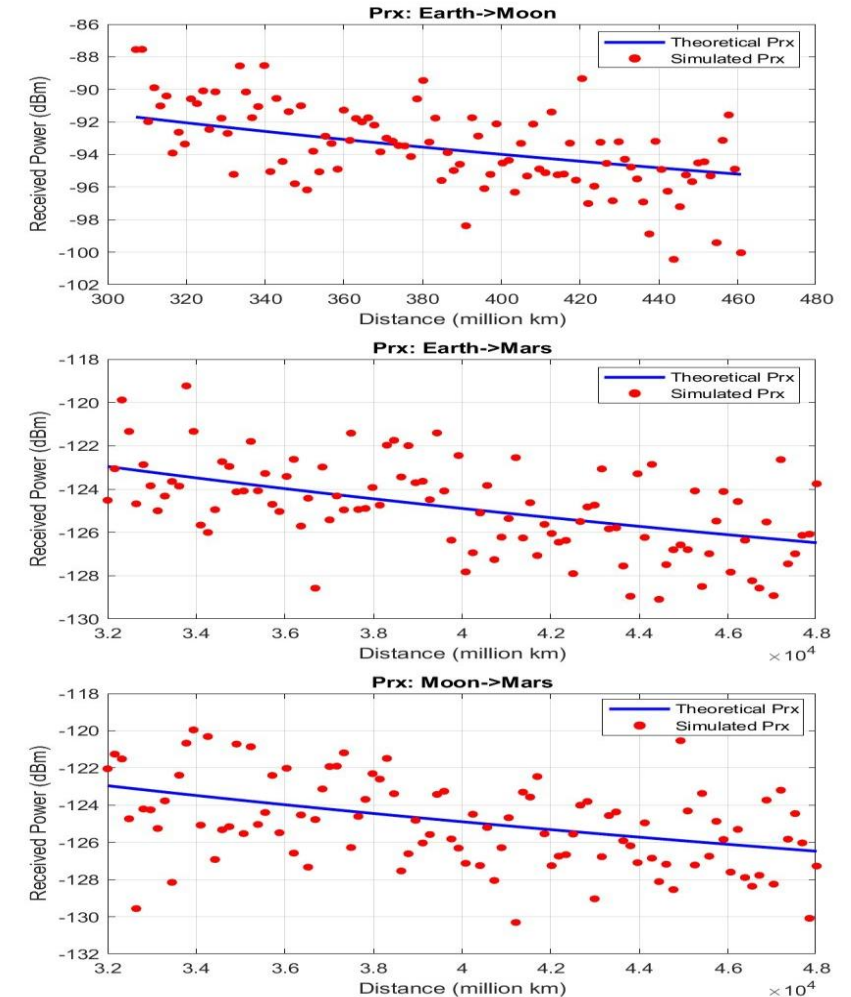


Fig. 3: Monte Carlo Scatter plot

Results & Discussion

- Histogram representations in Fig. 4 are used to illustrate the distribution of received power values.
- Theoretical curve aligns well with the peak of the histogram, confirming the accuracy of the link budget model.
- Case A: Highly stable with minimal variations.
- Case B: Received power ranges from -140dBm to -115 dBm.
- Case C: Power varies between -135 dBm to -115 dBm.
- Theoretical and simulated results are within acceptable thresholds.

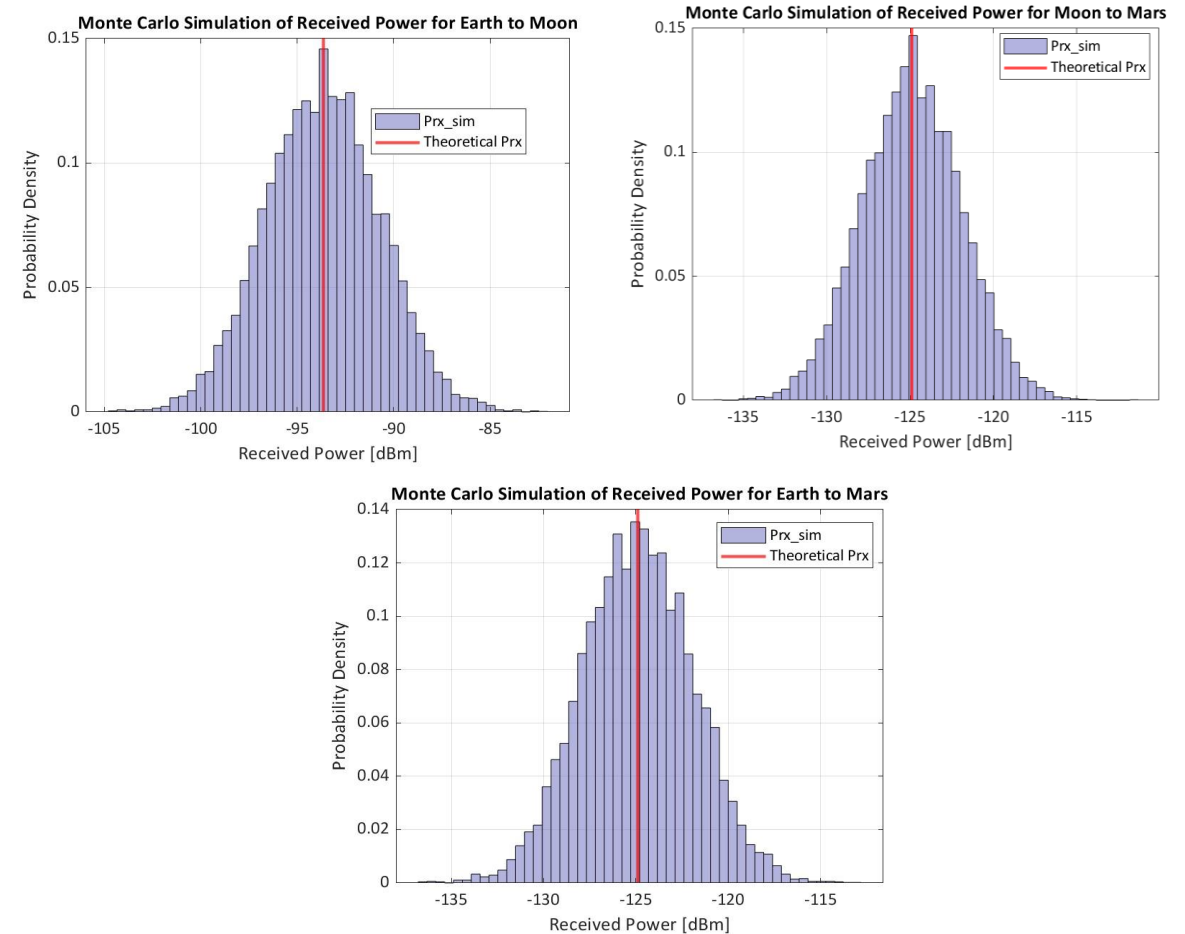


Fig. 4: Monte Carlo Histogram

SCPS/IPv6 Implementation

- SCPS (9-byte) encapsulated in IPv6 (40-byte)
- Successfully builds, parses and recovers **“Hello from deep space! This is a test message...”**
- **Packet Overhead Analysis (Table III):**
 16-byte payload (for example):
 SCPS: 25 bytes (36.0% overhead)
 IPv6: 65 bytes (61.5% overhead)
- Larger payloads reduce overhead, improving efficiency.
- SCPS/IPv6 is robust and optimized for deep space links.

Payload (Bytes)	SCPS Packet (Bytes)	IPv6 Packet (Bytes)	SCPS overhead (%)	IPv6 overhead (%)
16	25	65	36.0	61.5
64	73	113	12.3	35.4
256	256	305	3.4	13.1
1024	1033	1073	0.9	3.7
4069	4106	4145	0.2	1.0

Table III: SCPS/IPv6 overhead results

SCPS-TP vs LTP Performance

Throughput Comparison (Fig. 5):

SCPS-TP: 90,000-100,000 bytes/s, while LTP: 80,000 bytes/s. SCPS-TP more efficient for high-latency deep-space links.

Congestion Window Growth (Fig. 6):

SCPS-TP: 120,000 bytes at 70,000s while LTP: 100,000 bytes (20% smaller window). SCPS-TP utilizes available bandwidth more effectively.

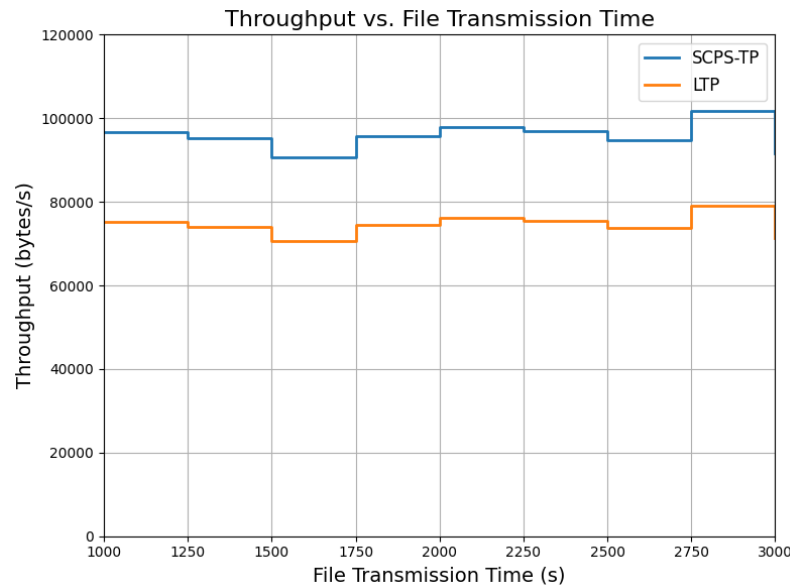


Fig. 5: Throughput vs File Transmission Time

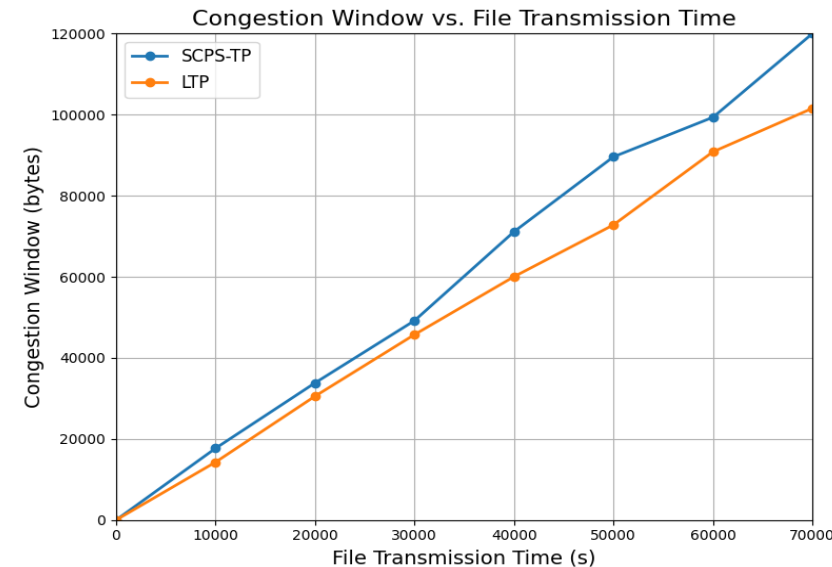


Fig. 6: Congestion Window vs File Transmission Time

Reinforcement Learning & Anomaly Detection

Reinforcement Learning for DTN:

- Modelled in Python, optimizes routing using a reward-penalty system.
- Optimal paths earn rewards; suboptimal paths incur penalties.
- Early training: Inefficient route, highly negative reward.
- Later episodes: Stabilized rewards.

Anomaly Detection with Isolation Forest:

- Detects outliers in communication delays.
- Identifies unusual spikes/dips in all 3 links.

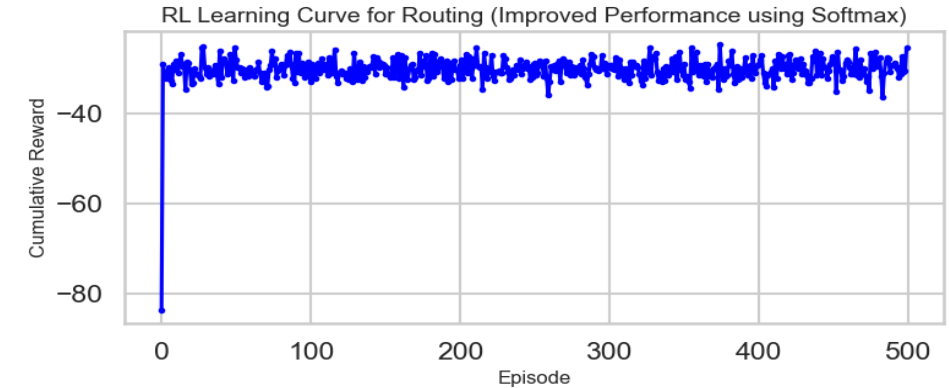


Fig. 7: Learning curve of RL based DTN

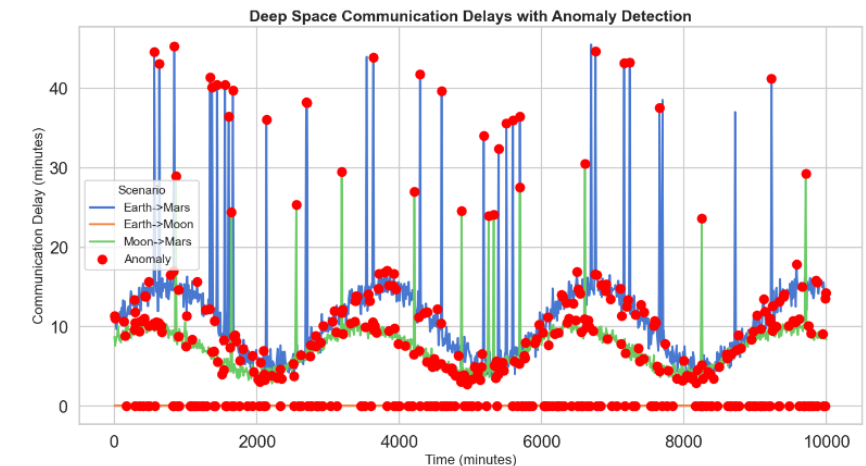


Fig. 8: Anomaly Detection in Communication delays

Comparison of DTN protocols

Packet Delivery Ratio (PDR) & Overhead Ratio Analysis:

- CGR: Contact graph routing
Maximum PDR, negligible overhead because of deterministic contact scheduling.
- CDT: Nearly optimal delivery at low overhead, needs ample training data.
- Epidemic: Minimum PDR, maximum overhead because of inefficient flooding.
- PRoPHET: Better than Epidemic but still has considerable overhead.
- RL-DTN: Balanced performance, better than Epidemic & PRoPHET in both areas.
- RL-DTN is efficient without rigid scheduling (CGR) or vast labelled data (CDT).

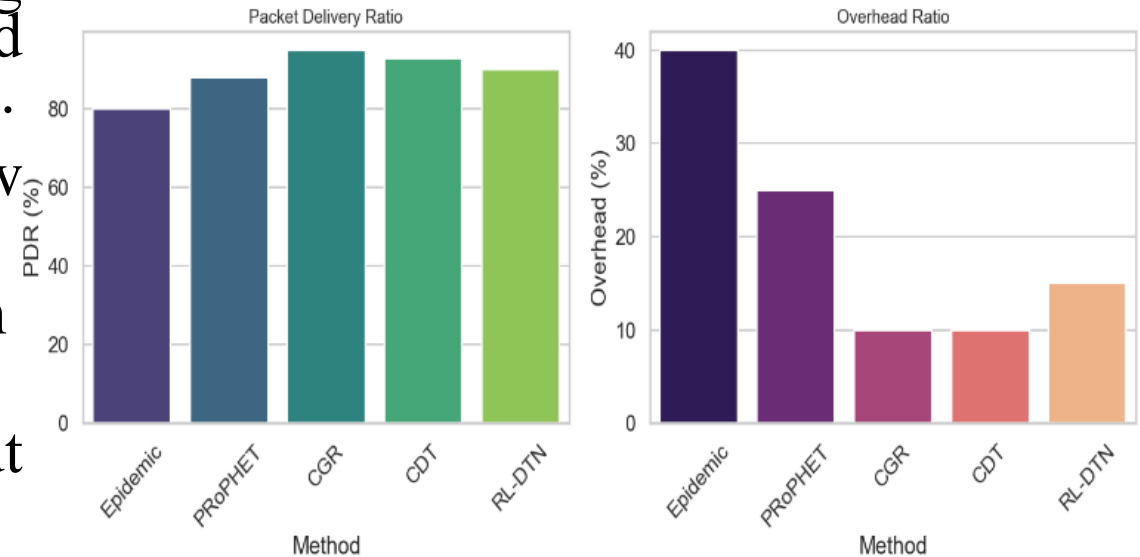


Fig. 9: Comparison of RL-DTN against existing models

Comparison of DTN protocols

Hamming Loss evaluates how often forwarding decisions are incorrect in DTNs.

Lower Hamming Loss = Smarter, More Efficient Routing

$$L_H(y, h(x)) = \frac{1}{m} \sum_{i=1}^m 1(y_i \neq h_i(x))$$

$$L_s(y, h(x)) = 1(y \neq (h(x)))$$

RL-DTN (Proposed) – Strong performance (0.22) via adaptive reinforcement learning.

Adaptive, learning-driven strategy can come surprisingly close to the performance of schedule-aware methods even without knowing future contact opportunities in advance.

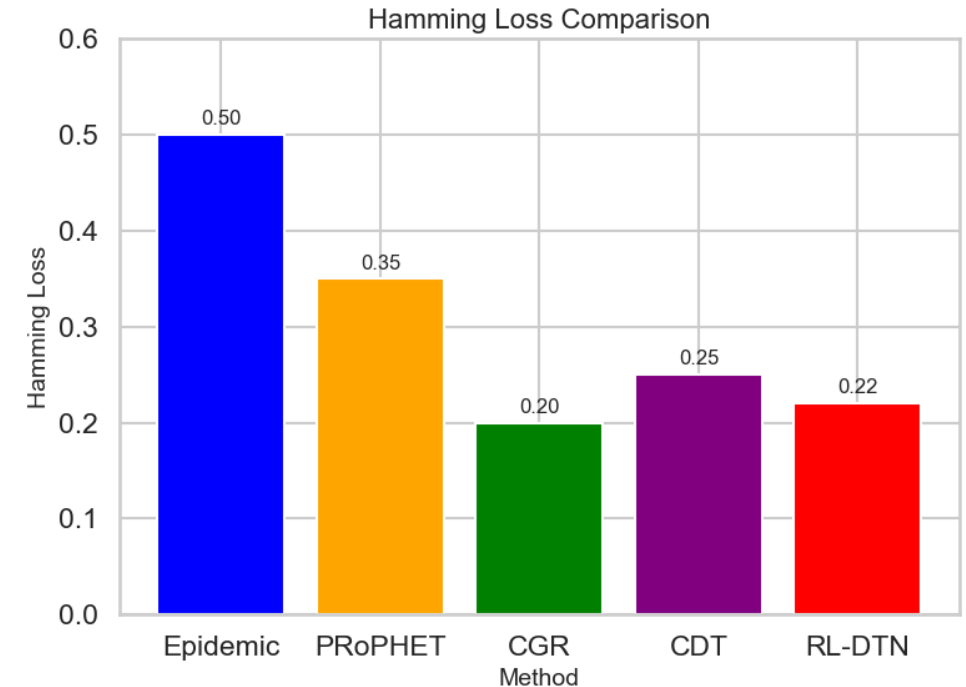


Fig. 9: Comparison of RL-DTN against existing models

Conclusion and Future work

This paper presents a comprehensive analysis of a Ka-Band and IP Stack based communication system which utilizes reinforcement learning and isolation forest algorithms for optimal routing and anomaly detection. The Monte Carlo simulations carried out on MATLAB prove the high gain antenna to be a suitable system. The RL model is proven to be able to train itself to find the most optimal routes. The isolation forest algorithm is successfully able to isolate and detection anomalies in the synthetic dataset. IPv6 and SCPS TP protocols were successfully demonstrated with SCPS-TP performing better than LTP in deep space conditions and RL-DTN is shown to be a balance between existing DTN protocols.

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